

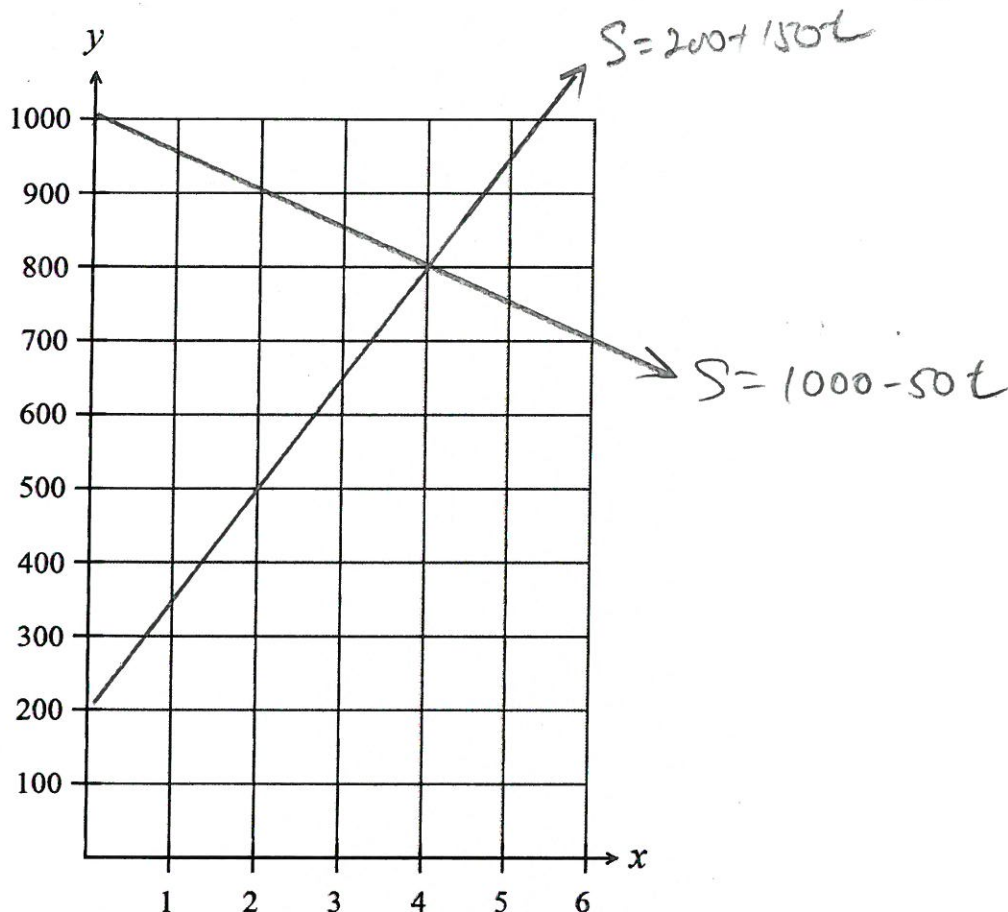
Question 4**(13 marks)**

(a) John had savings of \$200 and planned to save \$150 each week starting next week.

- (i) Determine an equation that gives the amount John had saved in terms of t where t is the number of weeks starting today at $t = 0$. (2)

$$S = 200 + 150t$$

- (ii) Sketch the equation on the set of axes below. (2)



At the same time, his brother Ian met a girl that intrigued him. He decided to spend \$50 a week on the girl from his savings of \$1000 and get to know her.

- (iii) Determine the equation that represents Ian's savings in terms of t . (2)

$$S = 1000 - 50t$$

- (iv) Sketch the equation that represents Ian's savings on the set of axes.
Label each graph. (2)

- (v) Determine when the two brothers have the same savings. (1)

week 4

- (vi) Verify this result algebraically. (2)

$$\begin{aligned}200 + 150t &= 1000 - 50t \\200t &= 800 \\t &= 4\end{aligned}$$

- (b) Solve for y

$$5y - (1 - 2y) + 7 = 20$$

(2)

$$5y - 1 + 2y + 7 = 20$$

$$7y + 6 = 20$$

$$7y = 14$$

$$y = 2$$